

Assignment - Day 1

Q1) a) Decimal - 143.75

Binary -

10001111.11 (2)

$$\begin{array}{r} 2 \overline{) 143} \\ 2 \overline{) 71} - 1 \\ 2 \overline{) 35} - 1 \\ 2 \overline{) 17} - 1 \\ 2 \overline{) 8} - 1 \\ 2 \overline{) 4} - 0 \\ 2 \overline{) 2} - 0 \\ \underline{1} - 0 \end{array}$$

$$0.75 \times 2 = 1.50 - 1$$

$$0.50 \times 2 = 1.00 - 1$$

Octal - 217.6 (8)

$$8 \overline{) 143}$$

$$0.75 \times 8 = 6$$

$$8 \overline{) 17} - 7$$

$$8 \overline{) 2} - 1$$

$$\underline{0} - 2$$

Hexadecimal -

$$16 \overline{) 143}$$

$$0.75 \times 16 = 12$$

$$16 \overline{) 8} - 15$$

$$\underline{0} - 8$$

$$= 815.12$$

8F.C

b) Binary - 11011.101

Decimal - $1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$$16 + 8 + 0 + 2 + 1 = 27$$

$$101 - 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} = 0.5 + 0 + 0.125 = 0.625$$

$$27 + 0.625 = 27.625$$

Octal - 011011.101

$$33.5$$

Hexadecimal - 11011.101

$$00011011.1010$$

$$\underline{1B.A}$$

c) Octal - 453

$$\text{Decimal} - 4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0 = 256 + 40 + 3 = 299$$

$$\text{Binary} - 453 = 100101011$$

$$\text{Hexadecimal} - 453 = 000100101011 = 12B$$

d) Hexadecimal - 1F.C

$$\text{Decimal} = 1 \times 16^1 + F \times 16^0 + C \times 16^{-1}$$

$$16 + 15 + 12 \times 0.0625$$

$$16 + 15 + 0.75 = \underline{\underline{31.75}}$$

$$\text{Binary} = 1FC \rightarrow 00011111.1100$$

$$\text{Octal} = 1FC \rightarrow 00011111.1100 \Rightarrow \underline{\underline{37.6}}$$

Q2) Find Equivalents:

a) $324_5 = ?_{10} \Rightarrow 3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0$

$$75 + 10 + 4 = 89_{10}$$

b) $32.23_4 = ?_{10} \Rightarrow 3 \times 4^1 + 2 \times 4^0 + 2 \times 4^{-1} + 3 \times 4^{-2}$

$$12 + 2 + 0.5 + 0.1875$$

$$= \underline{\underline{14.6875_{10}}}$$

c) $231_{10} = ?_7 \Rightarrow 7 \overline{) 231}$

$$7 \overline{) 33} - 0$$

$$7 \overline{) 4} - 5$$

$$0 - 4$$

$$450_7$$

d) $189_{10} = ?_6 \Rightarrow 6 \overline{) 189}$

$$6 \overline{) 31} - 3$$

$$6 \overline{) 5} - 1$$

$$0 - 5$$

$$513_6$$

Q3) Find the radix r :

a) $(312/20)_r = (13.1)_r$

$$\frac{312}{20} = 13.1$$

$$\Rightarrow \frac{3r^2 + r + 2}{2r} = r + 3 + \frac{1}{r}$$

$$312_r = 3r^2 + 1r + 2r^0 = 3r^2 + r + 2$$

$$20_r = 2r^1 + 0r^0 = 2r$$

$$13.1_r = 1r^1 + 3r^0 + 1r^{-1} = r + 3 + \frac{1}{r}$$

$$3r^2 + r + 2 = 2r \left(r + 3 + \frac{1}{r} \right)$$

$$3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$3r^2 - 2r^2 - 6r + r + 2 - 2 = 0$$

$$r^2 - 5r = 0$$

$$r(r - 5) = 0$$

$$r = 5$$

$$b) 264x + 136x = 433x$$

$$264x = 2x^2 + 6x + 4$$

$$136x = x^2 + 3x + 6$$

$$433x = 4x^2 + 3x + 3$$

$$(2x^2 + 6x + 4) + (x^2 + 3x + 6) = 4x^2 + 3x + 3$$

$$3x^2 + 9x + 10 = 4x^2 + 3x + 3$$

$$4x^2 - 3x^2 + 3x - 9x + 3 - 10 = 0$$

$$x^2 - 6x - 7 = 0$$

$$(x-7)(x+1) = 0$$

$$x = 7$$

$$c) 221505_8 = 12345_7$$

$$221505_8 = 2(8)^5 + 2(8)^4 + 1(8)^3 + 5(8)^2 + 0(8) + 5$$

$$= 65536 + 8192 + 512 + 320 + 5 = 74565$$

$$12345_7 = x^4 + 2x^3 + 3x^2 + 4x + 5$$

$$= 16^4 + 2(16)^3 + 3(16)^2 + 4(16) + 5$$

$$= 65536 + 8192 + 768 + 64 + 5$$

$$= 74565$$

$$x = 16$$

4) Addition:

$$a) 1101_2 + 11011111_2$$

$$\begin{array}{r} 11011111 \\ 1101 \\ \hline 11101100 \end{array}$$

$$b) 74652_8 + 76543_8$$

$$\begin{array}{r} 1111 \\ 74652 \\ 76543 \\ \hline 176415 \end{array}$$

$$c) 578A_{16} + A3D1E_{16}$$

$$A + E = 10 + 14 = 24 - 16 = 8$$

Result Carry

$$8 \quad 1$$

$$8 + D + 1 = 8 + 13 + 1 = 22 - 16 = 6$$

$$6 \quad 1$$

$$7 + B + 1 = 7 + 11 + 1 = 19 - 16 = 3$$

$$3 \quad 1$$

$$5 + A + 1 = 5 + 10 + 1 = 16 - 16 = 0$$

$$0 \quad 1$$

Carry

$$1$$

$$0$$

$$1$$

$$\Rightarrow \underline{10368}_{16}$$

$$d) \quad 234_6 + 315_6$$

$$4 + 5 = 9 \Rightarrow 1 + 6 \times 3 = 13$$

$$3 + 1 + 1 = 5 \Rightarrow = 805$$

$$2 + 3 = 5 \Rightarrow = 05$$

$$234$$

$$315$$

$$\underline{553}_6$$

Q 5) Binary $\rightarrow$ Gray code

$$a) \quad 10101_2 \rightarrow 1111$$

$$b) \quad 01110_2 \rightarrow 01001$$

Q 6) Gray code $\rightarrow$ Binary

$$a) \quad 01101 \rightarrow 01001$$

$$b) \quad 10101 \rightarrow 11001_2$$